



GSSS | GEETHA SHISHU
SHIKSHANA SANGHA(R)
**INSTITUTE OF ENGINEERING
AND TECHNOLOGY FOR WOMEN**

(Affiliated to Visvesvaraya Technological University, Belagavi, Approved by AICTE New Delhi & Govt of Karnataka)
KRS Road, Metagalli, Mysuru - 570016



12. Design and implement C/C++ Program for N Queen's problem using Backtracking.

```
#include<stdio.h>
#include<math.h>
#include<stdlib.h>

int board[20],count;

int main()
{
    int n,i,j;
    void queen(int row,int n);
    printf("-NQueensProblemUsingBacktracking-");
    printf("\n\nEnter number of Queens:"); scanf("%d",&n);
    queen(1,n);
    return 0;
}
```

//function for printing the solution

```
void print(int n)
{
    int i,j;
    printf("\n\nSolution%d:\n\n",++count);
    for(i=1;i<=n;++i)
        printf("\t%d",i);
    for(i=1;i<=n;++i)
    {
        printf("\n\n%d",i);
        for(j=1;j<=n;++j)//for nxn board
        {
            if(board[i]==j)
                printf("\tQ");//queen at i,j position
            else
                printf("\t-");//empty slot
        }
    }
}
```

/*function to check conflicts, If no conflict for desired position returns 1 otherwise returns 0*/

```
int place(int row,int column)
{
    int i;
```



```
for(i=1;i<=row-1;++i)
{
    //checking column and diagonal conflicts
    if(board[i]==column)
        return 0;
    else
        if(abs(board[i]-column)==abs(i-row))
            return 0;
}
return 1;    //noconflicts
}

//function to check for proper positioning of queen
Void queen(int row,int n)
{
    int column;
    for(column=1;column<=n;++column)
    {
        if(place(row,column))
        {
            board[row]=column;//noconflictssoplacequeen
            if(row==n)//deadend
                print(n);    //printingtheboardconfiguration
            else    //try queen with next position
                queen(row+1,n);
        }
    }
}
```

OUTPUT:

Enter number of Queens:4

Solution 1:

Solution 2:

	1	2	3	4	1	2	3	4
1	-	Q	-	-	1	-	-	Q
2	-	-	-	Q	2	Q	-	-
3	Q	-	-	-	3	-	-	Q
4	-	-	Q	-	4	-	Q	-